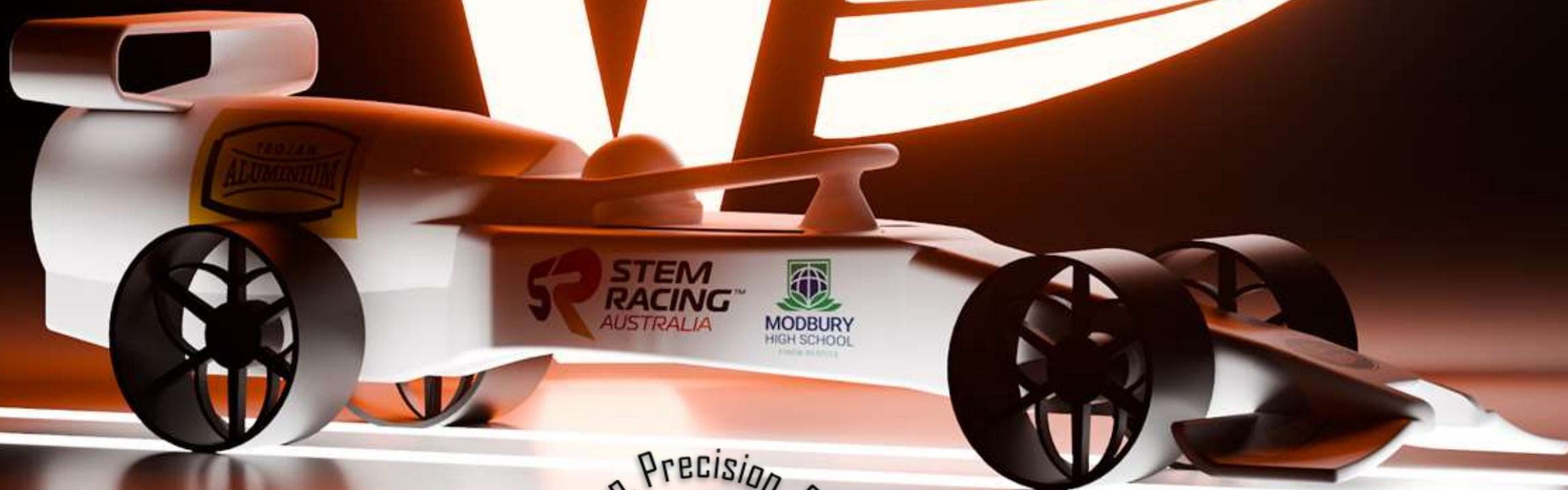


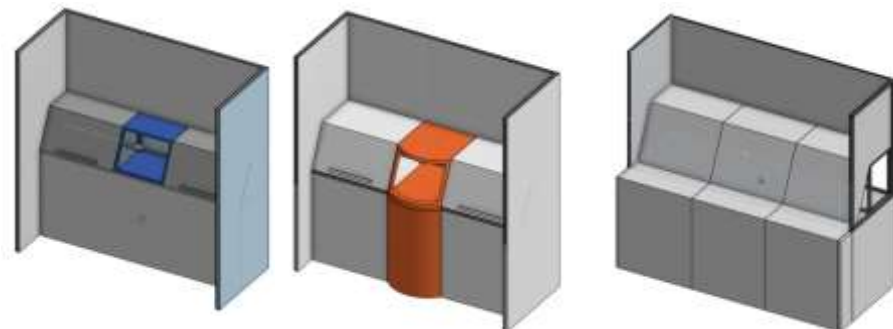
Velocity Vortex
Trade Display
Development Brief
National Final 2026

velocity vortex



Design Inspiration and Iterative Development: From Concept to Reality

The development of the Velocity Vortex Trade Display was more than a project—it was a transformative journey reflecting our team’s commitment to collaboration, innovation, and sustainability. Beyond meeting competition requirements; we aimed to redefine standards for visual appeal, functionality, and professionalism. This endeavour was deeply rooted in lessons learned from our State Final display, which served as a catalyst for improvement. By critically analysing what worked and what didn’t, elevated every aspect of our new design to not only meet but exceed expectations. Our initial concept featured a simple integrated car display but was considered too simplistic. A second concept introduced a curved car display, however this created unnecessary complexity. Our final design refined and combined the strongest elements of previous concepts, integrating a trestle table instead of a custom stand, and introducing the use of corflute strips, as pictured below.



The three major versions from left to right. (V1, V2, FINAL V9)

Credit - Riley Hutchinson

The Brainstorming Phase: Identifying Flaws and Envisioning Solutions

The first step in our process was a comprehensive brainstorming session, where we dissected the flaws of our previous Trade Display. This was not just a casual discussion—it was a structured analysis aimed at identifying structural weaknesses, inefficiencies in space utilisation, and inconsistencies in visual presentation. Among the key issues we identified were:

- **Structural instability**, which had led to wobbling and misalignment during setup.
- **Inefficient use of space**, resulting in a cluttered appearance that detracted from the car’s prominence.
- **Lack of visual cohesion**, where branding elements felt a bit disjointed instead of unified.

By pinpointing these weaknesses, we were able to develop targeted strategies to address them. For example, we recognised that the frame needed reinforcement to ensure stability, the layout required optimisation to maximise space, and the branding needed a more integrated approach to create a seamless visual experience. This phase was critical in shaping our vision for the new display, as it allowed us to focus our efforts on the areas that would yield the most significant improvements.

Initiation Process / Project Schedule / Resource & Budget / Team, Tasks & Responsibilities / Sponsorship ROIs & Comms / Team Identity / Social Media / Trade Display / Risks, Monitoring / Future Skills

The Role of Onshape in Collaborative Design

With a clear understanding of what needed to be improved, we turned our attention to design execution. Riley took the lead in modelling the trade display using Onshape, a browser-based CAD tool that proved to be a game-changer for our team. Unlike traditional CAD software, which often presented technical challenges and accessibility issues, Onshape offered several distinct advantages:

- **Real-time collaboration:** Every team member could access, review the design from any device with an internet connection. This eliminated the need for messy file transfers and version control issues.
- **Cloud-based storage:** All iterations were automatically saved and versioned, allowing us to track changes and revert to previous designs if necessary.

Riley’s work in Onshape was meticulous and iterative, involving twenty different iterations of the design. Of these, three represented major overhauls that fundamentally reshaped the display’s structure and layout. Each major iteration was informed by team discussions centred around three broad core principles:

1. **Aesthetics:** We wanted the display to be visually striking, with a cohesive colour scheme, clean lines, and a professional appearance that aligned with our team’s branding.
2. **Functionality:** The layout needed to be optimised for ease of assembly, transport, and presentation, ensuring that the display could be set up quickly and efficiently during competitions.
3. **Feasibility:** We had to balance cost, material availability, and construction complexity, ensuring that our design was both practical and achievable within our constraints.

For minor iterations, one or two team members would propose an idea to Riley, discuss how it aligned with the criteria, and refine it through further iterations. This process allowed us to quickly converge on our final design that met our goals, which the team is very pleased with.

After months of refinement, we finalised a design that reused the frame from our previous display—a decision that reduced waste and cost, while incorporating a custom-designed shelf frame and Corflute strips. This combination created a cohesive, effective, and problem-free layout that addressed all the flaws we had identified in our earlier designs.

Display Layout: Showcasing Innovation Through Design

The Evolution of the Car Display Concept: From Centrepiece to Integrated Showcase

Our initial vision for the Trade Display was to position the car as the undeniable centrepiece, drawing immediate attention to what is arguably our most important asset. However, as the shelf frame design began to take shape, we quickly realised that a central car placement would be impractical due to several factors:

- **Structural constraints:** The shelf frame design made it difficult to securely position the car in the centre without compromising stability.
- **Space limitations:** A central placement would have crowded the display, making it harder to include other essential elements like informational panels and sponsor logos.
- **Visual balance:** We wanted the car to be a focal point, but not at the expense of the display’s overall harmony and professionalism.

Rather than forcing a design that didn’t work, we re-imagined the layout to ensure the car remained a dominant feature while maintaining a clean, professional, and visually balanced arrangement. This involved creating a dedicated space for the car that was both prominent and integrated, ensuring it complemented rather than competed with other elements.

Strategic Placement of Branding and Information: Crafting a Professional and Engaging Experience

To enhance the Trade Display’s professionalism and innovation, we adopted a strategic approach to the placement of branding and informational elements. Our goal was to create a visually engaging experience that not only showcased the car but also told the story of our team’s journey. This involved:

- **Team Logo:** Centred and instantly eye-catching, the logo anchors our display, embodying our team’s identity. It’s bold, clean design ensures strong brand recognition from any angle.
- **Sponsor Logos:** Thoughtfully placed—visible yet unobtrusive—these logos honour our valued partners. Their presence enhances our professionalism, credibility, and the shared sense of achievement that defines our project.
- **Car Specifications and Design Process:** Clearly organised, this section outlines the car’s technical specifications, innovative design, and performance. It highlights our engineering expertise and problem-solving, inviting viewers to appreciate the precision behind every detail.
- **Team Member Roles and Contributions:** This section spotlights each member’s role and contributions, humanising the project. It emphasises the teamwork, passion, and collaboration that fuel our success, proving that every milestone is a collective triumph.
- **QR Code:** Easily accessible, the QR code connects our physical display to our digital world. It grants instant access to our social media and website, deepening engagement and fostering lasting connections with our journey.

The background of the display features our logo, team name, and slogan, reinforcing our identity and creating a unified visual theme. The Corflute placement was designed to allow for flexibility, such as adjusting the angle of the shelves to optimize visibility and flow. By reusing the previous frame, we not only promoted sustainability but also ensured structural integrity and resource efficiency, proving that innovation and responsibility can go hand in hand.

Modularity and Adaptability: Designing for Efficiency and Longevity

The Challenge of the 20-Minute Setup: Prioritizing Speed Without Sacrificing Quality

One of the most daunting challenges in Trade Display design is the 20-minute setup time limit imposed by competition rules. This constraint required us to prioritize modularity and ease of assembly without compromising the display's stability or visual impact. To achieve this, we focused on several key strategies:

- **Modular Components:** The display was broken down into manageable sections—such as the frame, shelves, and panels—that could be quickly assembled and secured. This modular approach not only simplified setup but also made the display easier to transport and store.
- **Reusable Steel Frame:** By reusing the steel frame from our previous display, we ensured a sturdy foundation that could be easily reconstructed. The frame's adjustable nature allowed us to fine-tune the layout as needed, providing flexibility without sacrificing stability.
- **Corflute Strips for Lightweight Strength:** Corflute was chosen for its lightweight yet durable properties, making it ideal for creating strong, adaptable panels that could be quickly slotted into place. Its versatility also allowed us to experiment with different angles and configurations during the design phase.

Testing and Refinement: Ensuring Flawless Execution

To guarantee that the display could be assembled efficiently and reliably, we conducted multiple setup tests under simulated competition conditions. These tests were invaluable in identifying and addressing potential issues, such as:

- **Misaligned Panels:** Early tests revealed that some panels were not fitting together seamlessly, leading to gaps and instability. We adjusted the measurements and reinforced the connections to ensure a snug, secure fit.
- **Unstable Connections:** The initial design had some weak points where components were prone to shifting during transport. We reinforced these areas with additional brackets and supports, ensuring the display remained rigid and reliable.
- **Lighting and Electrical Integration:** The RGB LED lighting had to be easily installable and securely mounted to avoid last-minute complications. We developed a plug-and-play system that allowed the lights to be quickly attached and powered by portable batteries, complying with competition rules.

These tests refined our assembly process, allowing us to streamline each step and ensure that the display could be constructed within the 20-minute limit. The modular nature of the design also means it can be easily re-configured for different events or future competitions, ensuring long-term usability and adaptability.

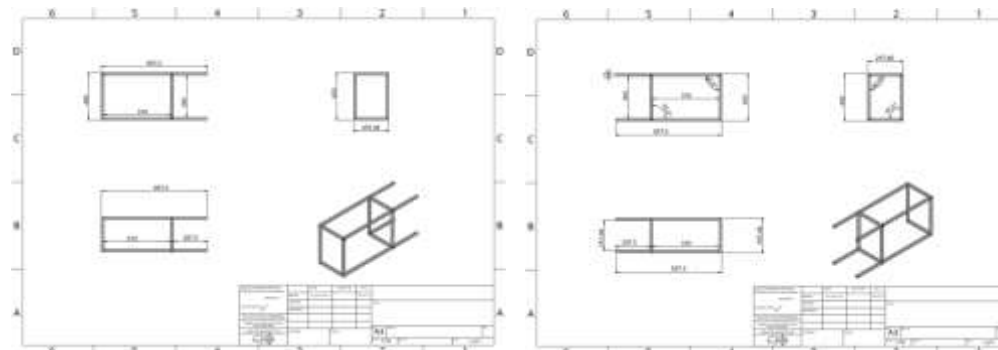
Compliance and Sustainability: Balancing Regulations and Responsibility

Maximizing Resource Efficiency: A Commitment to Sustainability

Sustainability was not just an afterthought—it was a core principle that guided every decision in our design process. We were deeply committed to minimizing waste and reducing our environmental impact, which led us to repurpose as many components as possible from our previous display. This included:

- **Trestle Table:** Serving as a stable base for the display, the trestle table was reused without modification, saving both materials and costs.
- **Steel Frame:** Instead of discarding the old frame, we reinforced and repurposed it, ensuring it could support the new design while maintaining structural integrity. This decision reduced material waste and lowered our carbon footprint.

By reusing these elements, we significantly reduced costs and minimized our environmental impact, aligning with our commitment to responsible resource use. This approach also demonstrated to judges and spectators that sustainability and innovation are not mutually exclusive—rather, they can enhance each other when thoughtfully integrated.



Trade Display Shelf Frame Technical Drawings (Left and Right Segments) Credit: Riley.H

Material Selection: Balancing Cost, Durability, and Environmental Impact

Selecting the right materials was a complex balancing act that required us to consider cost, durability, and sustainability. While we explored more eco-friendly alternatives such as Ecoboard, we ultimately chose Corflute for several compelling reasons:

- **Cost-Effectiveness:** Corflute was significantly cheaper than Ecoboard, allowing us to allocate our budget more efficiently to other critical components of the display.
- **Familiarity and Ease of Use:** Having worked with Corflute in previous projects, we were confident in its durability and versatility. This familiarity reduced the learning curve and allowed us to optimize its use for both structural support and visual appeal.

- **Lightweight and Adaptable:** The lightweight nature of Corflute made it easy to transport and assemble, while its adaptability allowed us to create custom shapes and angles that enhanced the display's aesthetics.

The steel frame was another critical material choice. While aluminum would have been preferable due to its lightweight and corrosion-resistant properties, we opted for steel because:

- **Availability:** Steel was readily accessible in our Tech Studies building, eliminating the need for additional purchases and reducing lead times.
- **Cost Savings:** Aluminum would have increased our overall costs, which was not feasible given our budget constraints. Steel provided a cost-effective alternative without compromising structural integrity.

In addition to these material choices, we repurposed tablets from a previous team, further reducing e-waste and promoting sustainability. This decision not only aligned with our environmental values but also demonstrated resourcefulness, a quality that judges often look for in competition entries.

Ensuring Compliance with Competition Regulations: Avoiding Pitfalls and Penalties

A critical aspect of the design process was ensuring that the display fully complied with all competition regulations. Failure to adhere to these rules could result in penalties or even disqualification, so we prioritized compliance from the very beginning. Key considerations included:

- **Size and Weight Restrictions:** The display had to fit within specific dimensions and be transportable in standardized boxes. We avoided monolithic or overly complex structures, opting instead for a modular, streamlined design that could be easily assembled, disassembled and packed.
- **Material Safety:** All materials used had to meet safety and environmental standards. We verified the compliance of every component, from the steel frame to the Corflute panels, ensuring they were non-toxic, durable, and competition-approved.
- **Electrical Regulations:** Since we incorporated RGB LED lighting, we had to ensure that the power source was compliant with competition rules, which prohibit plugging into external outlets. Our solution was to use battery-powered lights, which were safe, portable, and easy to install.

By proactively addressing these requirements, we eliminated the risk of non-compliance, ensuring that our display would be judged on its merits rather than penalized for technicalities. This attention to detail also simplified logistics, as the display could be quickly packed and transported without the need for specialized equipment or complex packaging solutions.

Aesthetic and Branding Choices: Crafting a Visual Identity

The Power of Colour: Creating a Striking and Cohesive Palette

The colour scheme of our Trade Display—black, white, and orange—was carefully selected to create a visually striking yet professional appearance. Each colour was chosen for its specific role in the overall design:

- **Black and White:** These colours provide a clean, neutral base that ensures the display remains versatile and adaptable. Black conveys sophistication and modernity, while white adds clarity and balance, making the display feel open and inviting. Together, they create a timeless, professional backdrop that allows other elements to stand out.
- **Orange:** was used as a vibrant accent color to highlight key features like the car, logo, and informational panels. It symbolizes energy, creativity, and boldness, reflecting our team’s dynamic and innovative spirit. The contrast between the neutral base and bold orange creates a visually engaging experience that captures attention.

This thoughtful colour palette not only enhances the display’s aesthetic appeal but also reinforces our team’s identity, making it instantly recognisable in a competitive setting.

Integrating Team Identity: From Logo to Slogan

The Velocity Vortex name, logo, and slogan were seamlessly integrated throughout the display to create a unified visual language that reflects our team’s values and aspirations. Here’s how we achieved this:

- **Logo:** Our logo was strategically placed in multiple locations, ensuring it is visible from all angles. The design itself incorporates elements of motion and precision, symbolising our commitment to speed, innovation, and technical excellence.
- **Team Name and Slogan:** The Velocity Vortex name and slogan were prominently displayed, reinforcing our brand identity and mission. The slogan, in particular, was crafted to embody our team’s ethos.
- **Consistent Branding:** The same colour scheme, fonts, and design elements were used across all materials—from the display itself to our brochures, website, and social media. This consistency ensures that our brand is cohesive and memorable.

By integrating these elements, we created a display that is not just a showcase for our car, but a powerful representation of who we are as a team—innovative, determined, and united in our pursuit of success.

Team Collaboration, Budget Management, and Lessons Learned

Defining Roles and Responsibilities: A Well-Oiled Machine

The development of the Trade Display was a true team effort, with each member playing a critical and well-defined role. This division of labour ensured that every aspect of the project was handled efficiently and effectively:

- **Riley’s Contributions:** Riley led CAD modeling using Onshape, creating a precise digital display design after 20 iterations. He also coordinated material sourcing, balancing aesthetics, functionality, and feasibility.
- **Vivaan’s Contributions:** Vivaan managed the budget and material procurement, repurposing old components to reduce costs and promote sustainability. He collaborated with the Tech Studies division to secure necessary equipment, keeping the project within budget while maximizing quality.
- **Team Effort:** The assembly phase was a collaborative effort, with all team members contributing to construction, testing, and refinement. This provided valuable experience in time management and adaptability for the upcoming competition.

Key Lessons and Takeaways: Growth Through Challenge

The journey of creating the Trade Display was not without its challenges, but each obstacle presented an opportunity for growth. Here are some of the most valuable lessons we learned along the way:

- **The Importance of Realistic Timelines:** Early in the project, we underestimated the time required for tasks, leading to delays and stress. This taught us the value of buffer time which ensured that we accounted for setbacks and maintained a realistic schedule.
- **Adaptability as a Core Skill:** When unexpected issues arose—such as material shortages, or software glitches—we learned to quickly consider options and alter our approach. This flexibility was key in keeping the project on track and maintaining momentum.
- **The Importance of Contingency Planning:** Frequent unexpected changes, such as last-minute material or design adjustments, highlighted the value of backup plans and agility. This proactive approach enabled us to overcome challenges while maintaining quality and meeting deadlines.

Budget

Our Trade Display cost was initially estimated at \$1000, but by repurposing materials from previous teams and our State Final display, we reduced the cost to approximately \$650 (not 100% final), creating an eye-catching and sustainable display that highlights our team’s dedication.

Final Outcome: A Display That Embodies Our Journey, Values, and Aspirations

Our Trade Display is far more than a competition requirement—it is a living testament to the creativity, dedication, and collective effort of every team member. It tells of our journey, from the initial brainstorming to the State Final and beyond, capturing the essence of our growth, resilience, and passion for engineering.

A Showcase of Innovation and Professionalism

Through the use of cost-effective materials, sustainable practices, and cohesive branding, we have created a display that is:

- **Fully Compliant:** Adhering to all competition regulations, ensuring that we avoid penalties and focus on performance.
- **Highly Functional:** With a modular and adaptable structure, the display is easy to assemble, transport, and reconfigure, making it versatile for future use.
- **Visually Stunning:** The bold colour scheme, professional layout, and integrated branding create a memorable and impactful presentation that stands out among competitors.

A Reflection of Our Team’s Spirit and Achievements

This display is not just a showcase for our car—it is a symbol of our team’s identity, representing our commitment to excellence, innovation, and collaboration. We are incredibly proud of what we have accomplished, and we believe this display exemplifies the best of Velocity Vortex:

- **Ingenuity:** Our ability to solve problems creatively and push the boundaries of what’s possible.
- **Teamwork:** The collective effort that went into every aspect of the project, from design to assembly.
- **Passion:** The dedication and enthusiasm that drives us to continuously improve and strive for greatness.

As we prepare to present this display at competitions, we do so with confidence and pride, knowing that it represents not just a project, but a journey—one defined by hard work, perseverance, and the unshakeable belief that great things are achieved through collaboration and innovation. This display is our story, and we are excited to share it with the world.